Rosenhain setup gate

Separate addendum · ROSENHAIN-SETUP-20261022

Models: 21; exact coefficient checks passed: 21.

Each model: 20 ordered finite-root pairs, 10 square scales.

Every chosen Fp map gives y'^2=t(t-1)(t-lambda)(t-mu)(t-nu).

First model p=739: r0=246, r1=369, scale=179, sqrt=87,

(lambda,mu,nu)=(296,16,10). Raw rows give all 21 maps.

Method: ordered root-pair search; square test; transformed quintic

reconstructed coefficient by coefficient and checked exactly.

The released binary was rerun byte-for-byte and a separate Rust

validator checked roots, square roots and coefficient vectors.

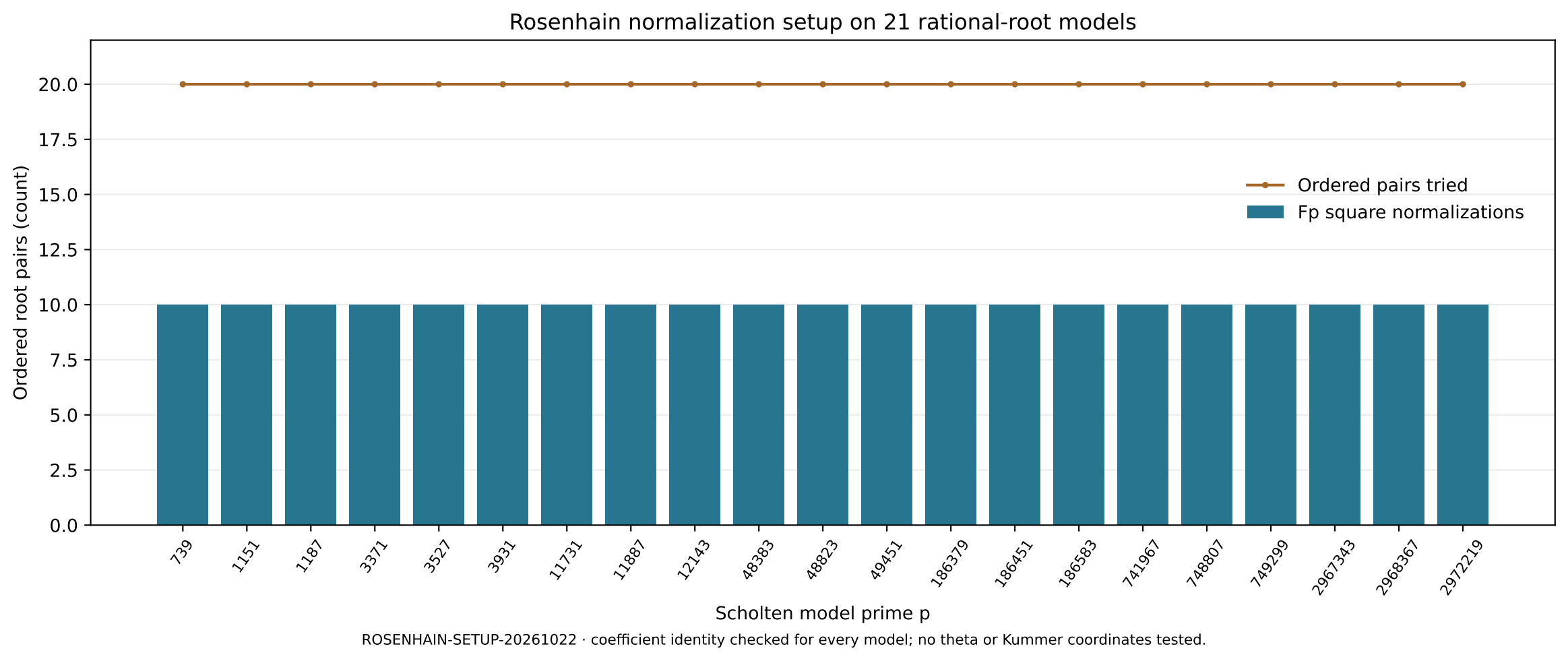
Boundary: theta constants, projection, xDBL/xADD, signed lifting,

and every candidate/rho price cell remain unmeasured.

Raw: results/rosenhain-models.jsonl and rosenhain-summary.json.

Primary formula context: https://eprint.iacr.org/2005/314/

Signed recovery context: https://eprint.iacr.org/2016/777/


Rosenhain normalization setup on 21 rational-root models
Ordered root pairs (count)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Ordered pairs tried
Fp square normalizations
739
1151
1187
3371
3527
3931
11731
11887
12143
48383
48823
49451
186379
186451
186583
741967
748807
749299
2967343
2968367
2972219
Scholten model prime p
ROSENHAIN-SETUP-20261022 · coefficient identity checked for every model; no theta or Kummer coordinates tested.